

$$1) P = T_2 \omega_2$$

$$T_2 = \frac{P}{\omega_2} = \frac{2.5 \times 10^3}{\frac{2\pi \times 1750}{60}}$$

$$= 13.64185227$$

$$\approx 13.64 \text{ Nm}$$

$$D_2 = m N_2$$

$$= 2.5 \times 20$$

$$= 50 \text{ mm}$$

$$F_t = \frac{2T_2}{D_2} = \frac{2(13.64)}{50 \times 10^{-3}}$$

$$= 545.6740906 \text{ N}$$

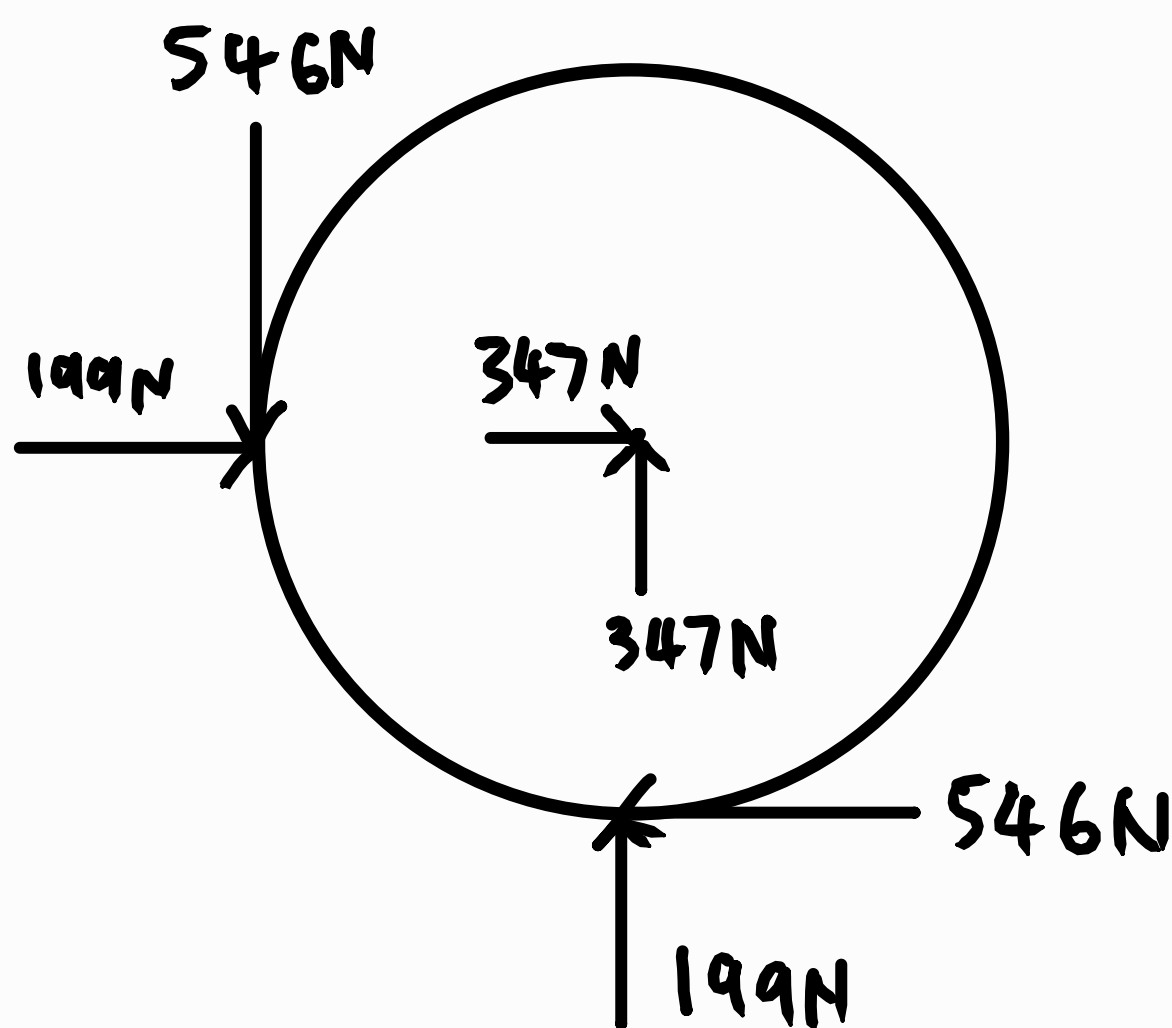
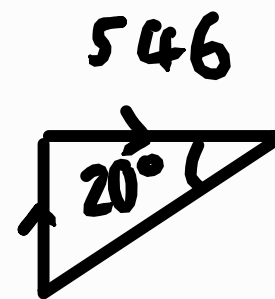
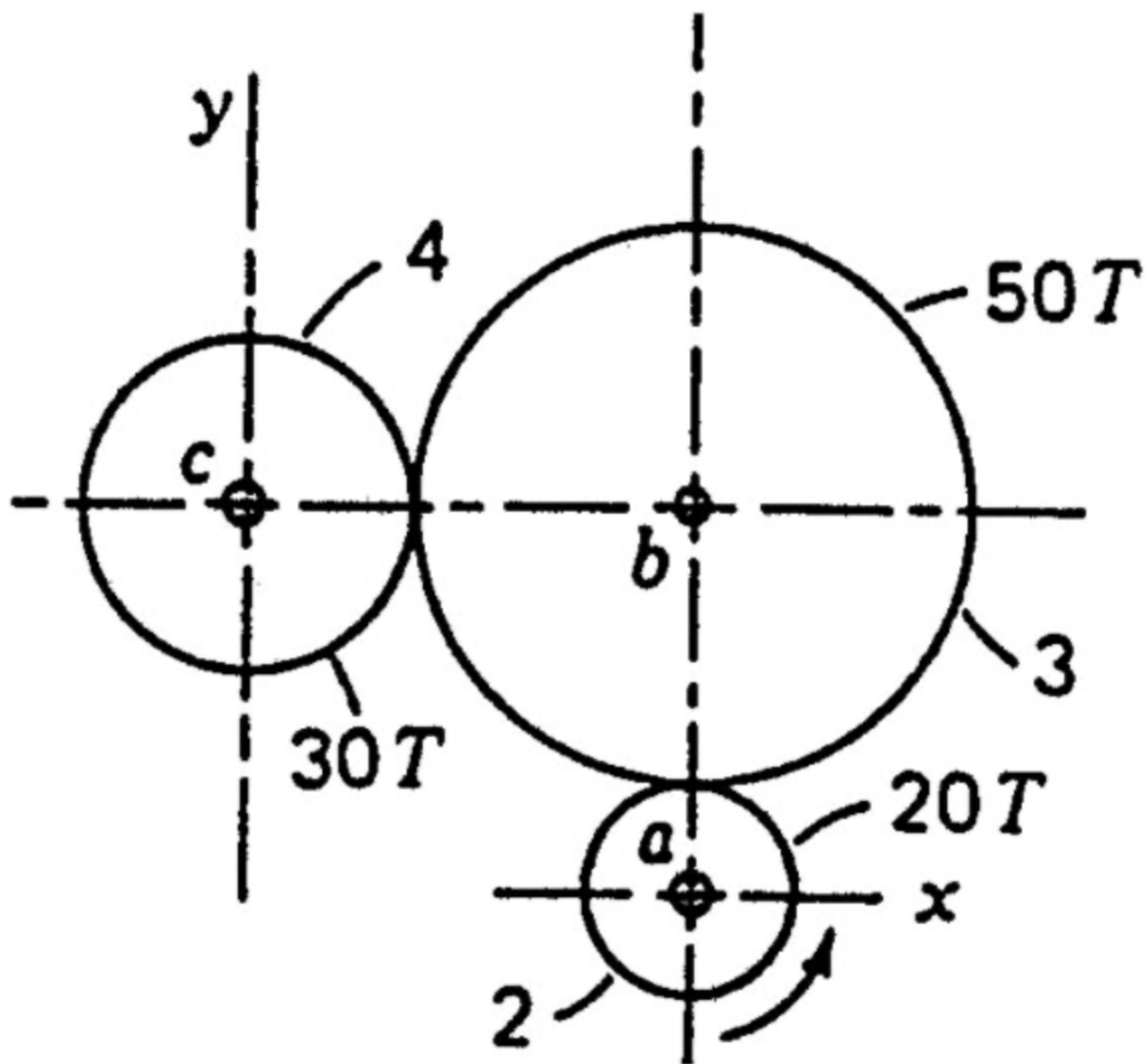
$$\approx 546 \text{ N}$$

$$\tan 20^\circ = \frac{546}{F_r}$$

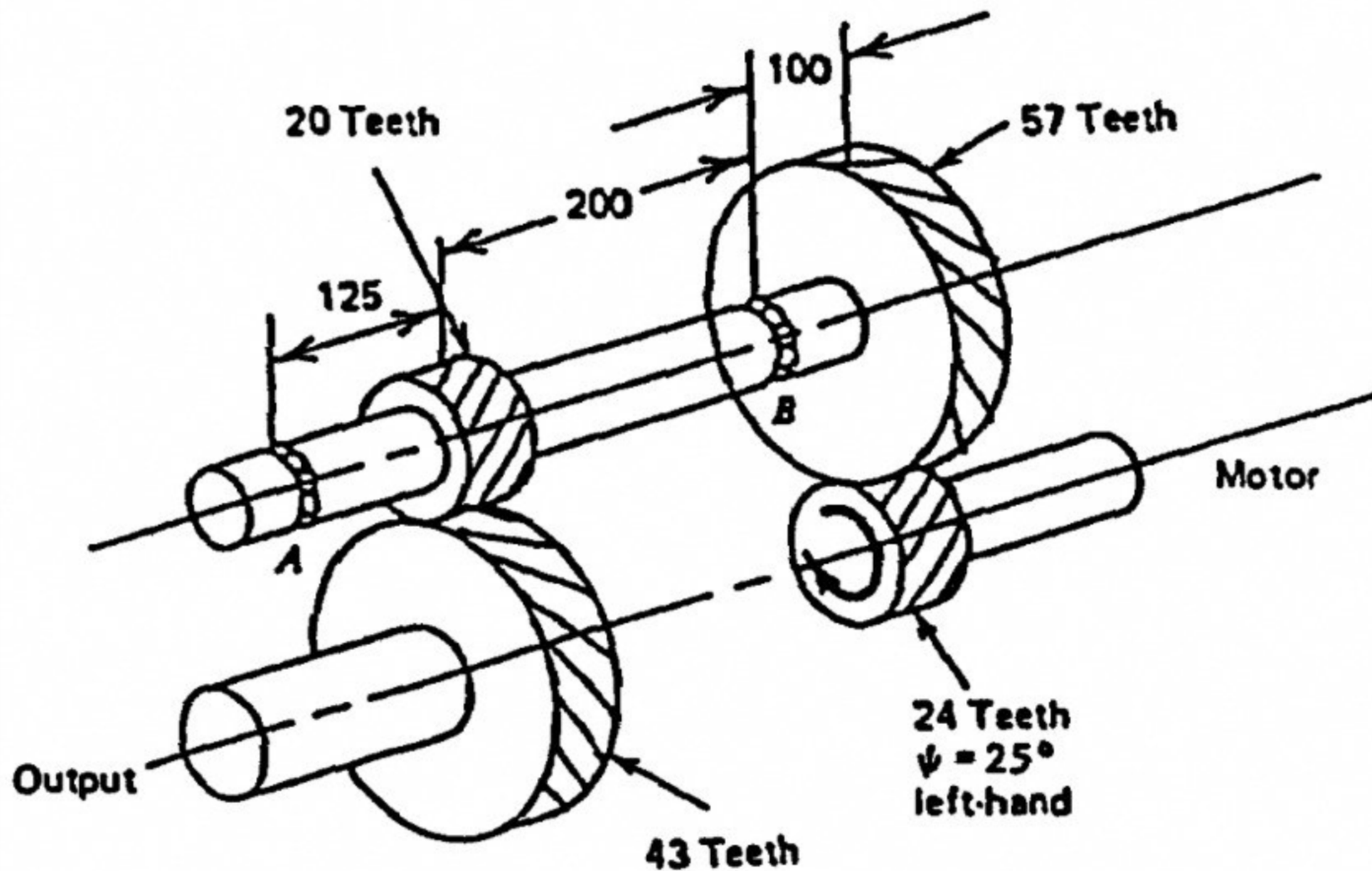
$$F_r = 545.7 \tan 20^\circ$$

$$= 198.6091266$$

$$\approx 199 \text{ N}$$



2)



a) $N_A = 20$ $N_B = 57$

The high-speed gears is on gear B.

For equal centre distances:

$$\frac{m(24+57)}{2} = \frac{m(20+43)}{2}$$

$$\frac{m_n}{\cos \psi} (40.5) = \frac{m_n}{\cos \psi} (31.5)$$

$$\frac{3}{\cos 25^\circ} (40.5) = \frac{4}{\cos \psi} (31.5)$$

$$\psi = 19.96946838^\circ \approx 19.97^\circ$$

2b) For the input gear:

$$\begin{aligned} D_i &= m N_i = \frac{m_n}{\cos \psi} N_i \\ &= \frac{3}{\cos 25^\circ} (24) \\ &= 79.44321017 \text{ mm} \\ &\approx 79.4 \text{ mm} \end{aligned}$$

$$\begin{aligned} T_i &= \frac{P}{\omega_i} = \frac{22 \times 10^3}{\frac{2\pi \times 600}{60}} \\ &= 350.1408748 \text{ Nm} \\ &\approx 350 \text{ Nm} \end{aligned}$$

$$\begin{aligned} F_{it} &= \frac{2T_i}{D_i} = \frac{2(350)}{79.4 \times 10^{-3}} \\ &= 8814.872261 \text{ N} \\ &\approx 8815 \text{ N} \end{aligned}$$

$$\begin{aligned} F_{ia} &= F_{it} \tan \psi = 8815 \tan(25^\circ) \\ &= 4110.44244 \text{ N} \\ &\approx 4110 \text{ N} \end{aligned}$$

$$\begin{aligned} F_{ir} &= F_{it} \tan \phi_t = F_{it} \frac{\tan \phi_n}{\cos \psi} = 8815 \left(\frac{\tan 20^\circ}{\cos 25^\circ} \right) \\ &= 3540.023784 \text{ N} \\ &\approx 3540 \text{ N} \end{aligned}$$

2b) For gear A:

$$D_A = m N_A = \frac{m_n}{\cos \psi} N_A$$

$$= \frac{4}{\cos 19.97^\circ} (20)$$

$$= 85.11801216 \text{ mm}$$

$$\approx 85.1 \text{ mm}$$

$$T_A = T_B = \frac{N_B}{N_i} T_i = \frac{57}{24} (350)$$

$$= 831.5845777 \text{ Nm}$$

$$\approx 831.6 \text{ Nm}$$

$$F_{A_t} = \frac{2 T_A}{D_A} = \frac{2 (831.6)}{85.1 \times 10^{-3}}$$

$$= 19539.56763 \text{ N}$$

$$\approx 19540 \text{ N}$$

$$F_{A_a} = F_A \tan \psi = 19540 \tan 19.97^\circ$$

$$= 7100.236491 \text{ N}$$

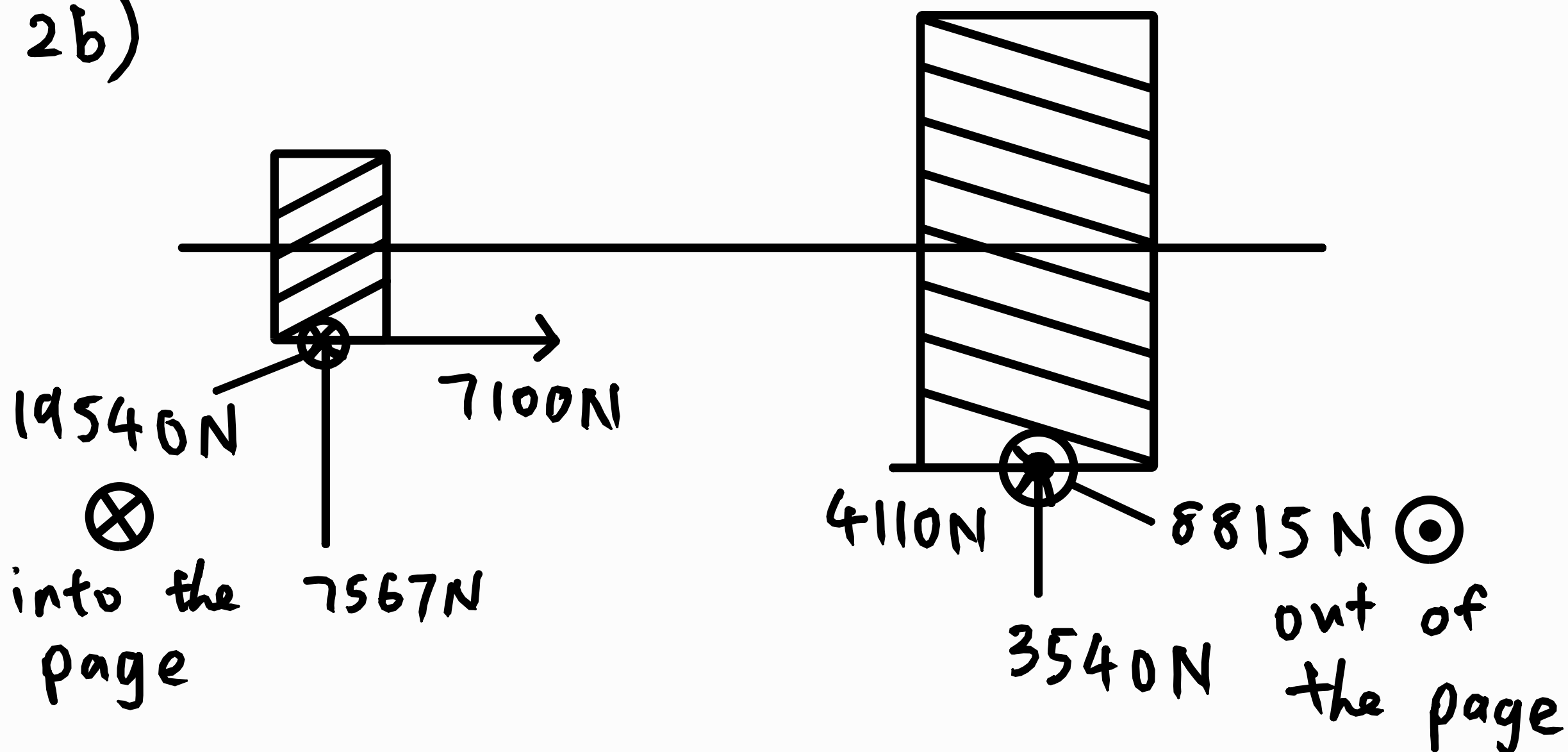
$$\approx 7100 \text{ N}$$

$$F_{A_r} = F_A \tan \phi_t = F_A \left(\frac{\tan \phi_n}{\cos \psi} \right) = 19540 \left(\frac{\tan 20^\circ}{\cos 19.97^\circ} \right)$$

$$= 7566.800839 \text{ N}$$

$$\approx 7567 \text{ N}$$

2b)



c) Net axial load:

$$F_{na} = 7100 + 4110$$

$$= 11210 \text{ N backwards}$$

d) The axial load on the 20-teeth gear will be in the opposite direction.

The new net axial load is:

$$F_{na} = 7100 - 4110$$

$$= 2990 \text{ N forwards}$$

The net axial load will be lower, at 2990 N, and its direction will change from backwards to forwards.